ECE551: Digital System Design and Synthesis

Fall 2026

Course Introduction



Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN–MADISON

Instructor

Julie Hsiao (julie.hsiao@wisc.edu)

- Assistant Teaching Professor
- Engineering Hall 3615
- Office hour: Thurs. 10:30-12:00pm



Education

- University of Toronto (BASc, MASc, PhD)

Research Interest

- Stochastic computing, approximate computing
- High-level synthesis, CAD
- Reconfigurable computing

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Teaching Team

TAs:

- Jiahao Lin
  - jlin445@wisc.edu



- Yucheng Huang
  - yhuang546@wisc.edu



Office hours

- Mon, Fri 12:05-12:55PM

ICPCs:

- Zenin Chen
- Srivibhav Jonnalagadda
- Ojas Patil

Grader:

- Shreya Somani

Class Format

Lectures

- Video lectures posted on Canvas → complete weekly quizzes
- MWF 9:55-10:45am in EH3654: hands-on exercises
- Many exercises work towards final project

Discussions

- Hosted by TAs
- Problem walk-through
- Optional but highly recommended!
- 2 sessions will be tool tutorial (held in B555)

Piazza

- General questions & discussion

Course Goals

- Learn how to use Hardware Description Languages (HDL) to design, verify, and realize digital logic
- Learn how to synthesize HDLs into hardware using the same tools used in industry
- Learn how to effectively work to design a digital system in a team environment

- Should be qualified for an entry-level job or internship at an IC design firm (AMD, ARM, Intel, Micron, Nvidia, Qualcomm, …) by the end

AI Use

- Allowed to use AI tools
- Transparency
  - Clearly state in submissions use OR non-use
  - State what it was used for
- Need to be able to explain all code
- We will have a small exercise on using AI for design / testing
  - After you know some basics
- You should be able to easily judge whether the AI tool is doing things right / well or not

What you should already know…

- Principles of basic digital logic design (ECE352)
  - Number representation (unsigned, signed, Hex & Binary)
  - Boolean algebra
  - Gate-level design
  - K-Map minimization
  - Finite State Machines
  - Basic datapath structures (adders, shifters, SRAM)
  - Some basic ModelSim and verilog from ECE352 AHWs
- How to log in to CAE machines and use a Linux shell

→ There is a Review Quiz that tests some of these basics
Please complete the Quiz by Friday (3 attempts)

Course Overview

- SystemVerilog basics – how to describe hw blocks
- Simulation Mechanics – making sense of simulation results
- Testbench constructs – create better testing infrastructure
- Synthesis flow
- Area/Delay Optimizations
- Physical Implementations

Course Goals

- Provide knowledge and experience in:
  - Digital circuit design using a HDL (Verilog/SystemVerilog)
  - HDL simulation
  - Good practices in digital design verification
    - How to build self-checking test benches
  - Synthesis of dataflow and behavioral designs
    - Optimizing hardware designs (timing, area, power)
  - Basic static timing analysis concepts
  - Introduce APR (Auto Place & Route)
  - Design tools commonly used in industry

- Teach you to think hardware first! …then code design in HDL

Course Materials

- Lecture slides
- No textbook necessary
- Standards
  - IEEE Std.1364-2001, IEEE Standard Verilog Hardware Description Language, IEEE, Inc., 2001.
  - IEEE Std 1364.1-2002, IEEE Standard for Verilog Register Transfer Level Synthesis, IEEE, Inc., 2002
- Tutorials on Canvas page for various CAD tools used
- Other useful readings

Evaluation and Grading

- 15% Homeworks } 15% penalty per 24hrs late, not accepted after 48hrs
- 10% Weekly Quizzes (lowest 1 dropped)
- 10% In-class exercise (lowest 2 dropped) } Need to be submitted on time to receive full grade
- 20% Project
- 20% Midterm
- 25% Final } Need to score >50% combined on exams to pass

Graded on a curve. Class average will be around 3.3

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Class Project

- Work in groups of 4
- Design, model, simulate, synthesize, and test a complex digital system
- One of the most important component of grade / class
  - In class exercises focus on building components needed
  - Homework problems that build toward the project
  - Exams will have some questions project related
  - Final code review and testing will be detailed
- More details coming later in the course



Course Tools

- Industry-standard design tools:
  - Modelsim / QuestaSim HDL Simulation Tools (Mentor / Siemens)
  - Design Vision / Design Compiler Synthesis Tools (Synopsys)

- Tutorials will be available for all tools
  - Modelsim tutorial next week (doc being updated)
  - Design Vision tutorial around week 8

Questions?



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In-Class Exercise

- Read the syllabus
- Complete EX00 quiz

Reminder for this week:

- Install / figure out access to tools
- Complete Review Quiz
- Watch week 1 lecture videos & complete quiz by Sunday